Rhaister

Back to basics: Observed statistics are sufficient to predict drug responses

Can summary statistics predict drug responses
in a new cellular context?

Valentine Svensson et al. · Tahoe Therapeutics
June 2026 preprint

Where this comes from: Tahoe’s atlas, Tahoe’s model

Tahoe Therapeutics built the perturbation atlas this field now trains on — and this paper is their own method on it

THE ATLAS · Feb 2025

Tahoe-100M

- 100M single-cell transcriptomes — the largest public perturbation atlas
- 50 cancer cell lines × 379 drugs × 3 doses, 24 h
- Mosaic “cell village”: the lines are pooled in one well and demultiplexed by genotype — batch effects die at the source
- 95.6M cells pass full QC · released free on HuggingFace

THE MODEL · Jun 2026

Rhaister / Rhaister-O

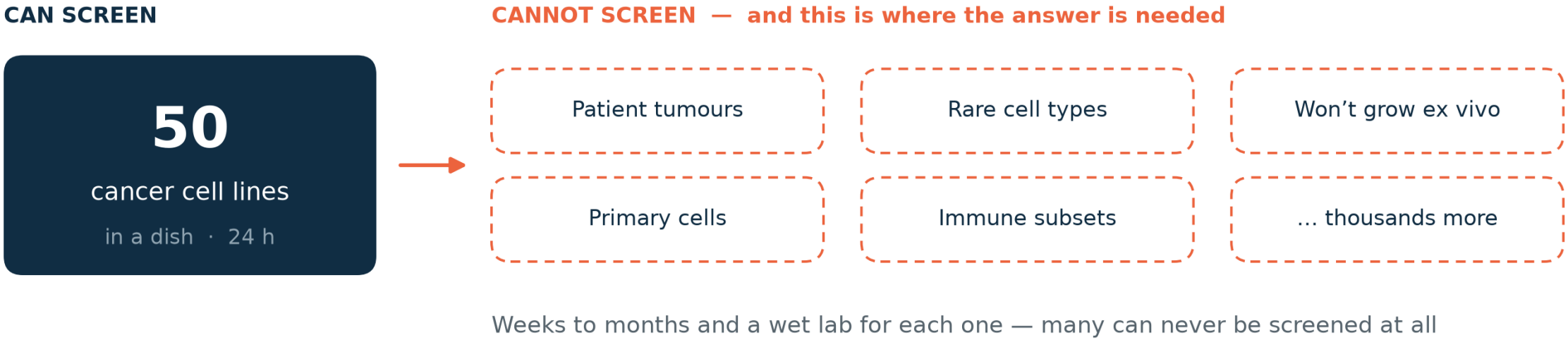
- Same group: 8 of its 10 authors wrote Tahoe-100M, including both of its corresponding authors
- So this is the data owner’s own method on their own benchmark — the reference point, not an outside challenger
- Weights and code on HuggingFace: tahoebio/Rhaister — seconds to train, milliseconds to predict
- We reproduced Rhaister-O at 0.6315 against the released ≈ 0.63

Lineage: Tahoe-100M (Feb 2025) → Tahoe-x1 foundation model (Oct 2025) → Rhaister (Jun 2026)

This is the benchmark’s owners telling you what a strong baseline looks like

Motivation: the hardest contexts can’t be screened

A screen reaches cell lines in a dish. The contexts that matter most are not in that set.

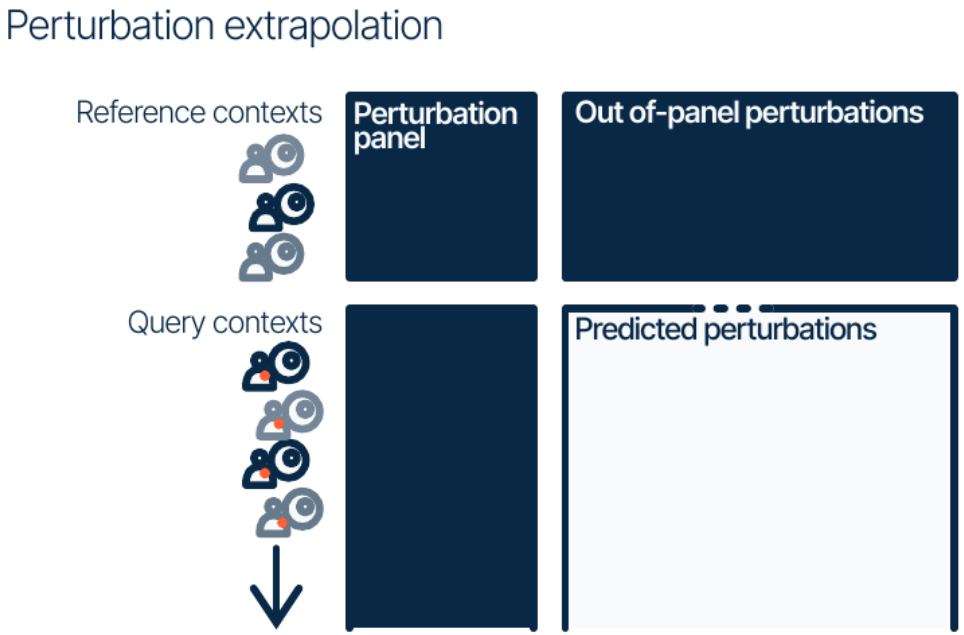


Today’s answer: virtual-cell models simulate single cells, then aggregate

Slow to train and run · no zero-shot capability · sometimes no better than simple baselines

→ Rhaister inverts it: predict the statistics directly, from baseline expression

The task: complete a drug screen in a new context



Known in reference cell lines

Responses to the full drug library

Measured in a new cell line

A panel of drug responses

Predicted

How new cell line responses to the drugs

Tahoe benchmark: 45 reference + 5 query cell lines; 120 panel + 240 target drugs

Svensson et al., 2026, Fig. 1 (top left), pp. 3–4 and 13

What do we have? The data and the notation

Every quantity below is a per-gene statistic over G genes — single cells are never modelled downstream

INPUT · context state

Baseline expression of cell line c

$$x_c = (x_{c,1}, \ldots, x_{c,G}) \in \mathbb{R}^G$$

- Pseudobulk over the untreated (DMSO) cells of line c
- Available for all 50 lines, treated or not
- The only query-line measurement Rhaister-O needs

TARGET · perturbation response

Response of line c to perturbation p

$$y_{c,p} = (y_{c,p,1}, \ldots, y_{c,p,G}) \in \mathbb{R}^G$$

- Treated vs. plate-matched DMSO, computed per line and plate
- p = (drug, dose): every dose is a separate perturbation
- Three views per gene: expression delta Δ, $\log_2$ fold change, MWU p-value

Tahoe-100M: 50 cancer cell lines × 384 drugs × 3 doses (0.05, 0.5, 5 µM), 24 h readout
Split: 45 reference / 5 query lines · 120 panel + 240 target drugs → 360 + 720 perturbations

Svensson et al., 2026, Data, pp. 13–14; summary statistics Eqs. 19–20, p. 14

Rhaister learns a weighted sum of panel responses

Fit the weights on the 45 reference lines, then apply them in the query line

1 · FIT — one ridge problem per target drug p★, solved across reference lines only

$$\widehat{W} = \arg\min_{W \in \mathbb{R}^{|D_x| \times G}} \sum_{c \in \mathcal{C}_{\text{ref}}} \| Y_c - W^\top X_c \|_2^2 + \lambda \|W\|_F^2$$

X = panel responses of the 45 reference lines (45 × |Dₓ| × G) · Y = their responses to the target drug p★ (45 × G) · λ = ridge penalty

2 · PREDICT — a weighted sum of the panel responses measured in the query line c★

$$\hat{y}^{\text{ridge}}_{c^\star, p^\star, g} = \sum_{q \in D_x} \widehat{W}_{q,g} \; y_{c^\star, q, g}$$

ŷ = the predicted statistic: one gene, one target drug, one query line.
y(c★, q, g) = the panel response already measured in the query line — the only query-line data Rhaister uses.

Fitted per view (delta, log₂FC, MWU), with an additive fallback and a small MWU calibration head

Svensson et al., 2026, p. 11, Eq. 3 (objective) and Eq. 4 (prediction); full framework Eqs. 1–7

Rhaister leads the Tahoe-100M comparison

Held-out cell lines, five splits — read the two highlighted columns

Model	Discrim.	Pearson Δ	PR-AUC	LFC rank	DE overlap	ES rank
Rhaister	**0.96**	**0.87**	**0.73**	**0.81**	**0.59**	**0.96**
STATE	0.93	**0.81**	**0.51**	0.57	0.40	0.89
Additive	0.82	**0.55**	**0.55**	0.61	0.43	0.85
Half-sample	0.93	**0.77**	**0.62**	0.67	0.52	0.97

Additive = cell-line effect + drug effect, no interaction. Half-sample = half the cells predicting the other half.
Not uniform elsewhere: on Replogle–Nadig, Additive and STATE beat Rhaister on LFC rank (0.66 / 0.78 vs 0.63).

The half-sample reference is an empirical comparison, not a hard ceiling.

Svensson et al., 2026, Fig. 1, Tahoe-100M panel; p. 12, half-sample reference

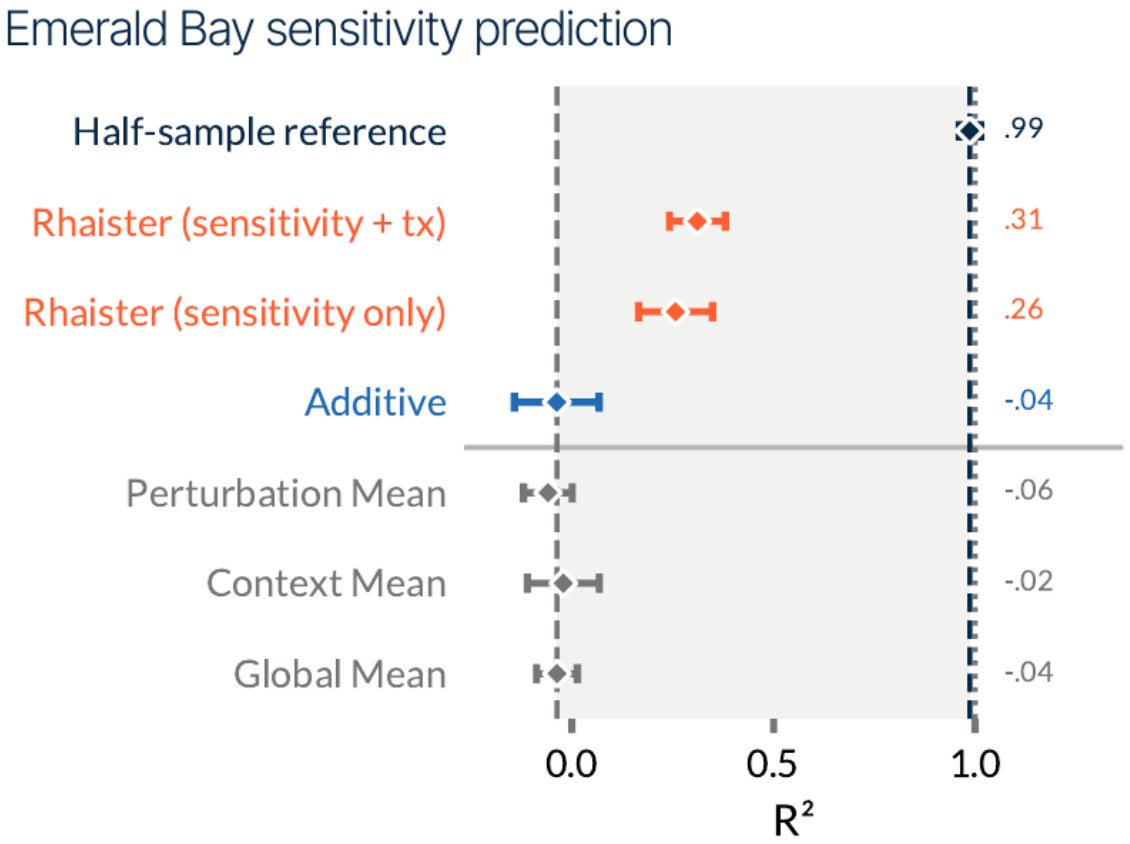
Transcriptomic responses improve sensitivity prediction

Emerald Bay

52 pooled cancer cell lines
5-day drug treatment
Paired abundance and RNA readouts

0.26 to 0.31

R^2 after adding transcriptomic responses from the measured panel



Endpoint: change in relative cell-line abundance within a pooled culture

Rhaister-O predicts from baseline expression alone

Zero-shot: no treated cells from the query line

1 · FIT — per target drug p, per gene g, over the 45 reference lines

$$\left(\widehat{\beta}_{p,g},\ \widehat{\gamma}_{p,g}\right) = \arg\min_{\beta,\gamma} \sum_{c \in \mathcal{C}_{\mathrm{ref}}} \left(y_{c,p,g} - \beta - \gamma\, x_{c,g}\right)^2$$

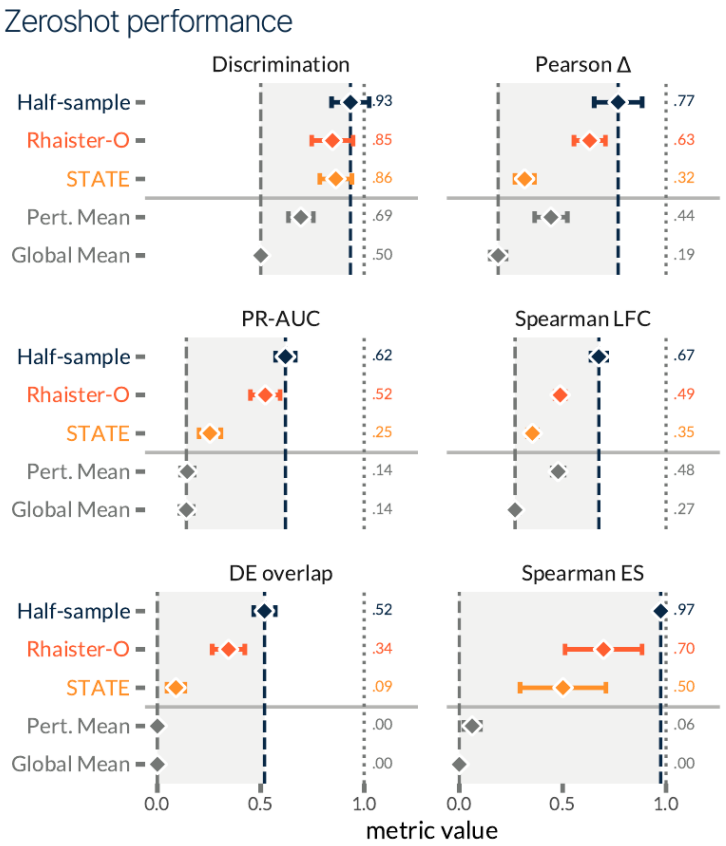
2 · PREDICT — query line c★, unseen target drug p★

$$\widehat{y}_{c^\star,p^\star,g} = \widehat{\beta}_{p^\star,g} + \widehat{\gamma}_{p^\star,g}\ x_{c^\star,g}$$

Two parameters per (drug, gene), fitted on 45 points — one per reference line.
x(c★, g) is the only query-line input, and it is untreated.

Pearson Δ: 0.63 vs 0.32

Rhaister-O compared with STATE
Discrimination: 0.85 vs 0.86



Svensson et al., 2026, Fig. 4 (top), p. 8; Eqs. 8–9, p. 11

Does it transfer? Our own check on L1000

Leave-one-cell-line-out · 8 LINCS L1000 lines, full 24 h coverage · 1,618 shared conditions · 949 landmarks · DepMap baseline

Line-macro Pearson	Tahoe (45 ref. lines)	L1000 (7 ref. lines)
Drug(-dose) mean	0.4434	0.2992
Rhaister-O	0.6315	0.2735
Gain from the treatment-specific slope	**+0.1881**	**−0.0257**
Same gain, drug-residual level	**+0.5380**	**+0.0453**

- Rhaister-O loses on 8 of 8 held-out lines, RMSE included; the bootstrap intervals do not overlap
- All four cell representations collapse onto the drug mean too (0.294–0.301 vs 0.300) — not a parameterisation problem

The estimator is not the point — Tahoe's condition-level reproducibility is.

Not apples to apples: Level-5 z-scores vs plate-matched deltas · 978 landmarks, only 79 shared with the Tahoe 2K panel · DepMap bulk baseline vs Tahoe DMSO. And 8 contexts against 45: the paper's own scaling test saturates near 20 reference contexts for the panel model, while baseline-only Rhaister-O keeps improving. Cohort size is a plausible partial cause — but "worse than doing nothing" is a different failure from "weak".

This project's own result, not Svensson et al. — Slurm jobs 274337 / 275031, 2026-09-23; report/09_25_report.md § 10

Summary statistics provide a strong baseline

The contribution

- Predicts the analysis-ready statistic directly, not simulated cells
- Rhaister-O: first zero-shot model here that clearly beats simple baselines

The boundary

- Known drug library, new cell contexts — not new chemistry
- Population summaries leave single-cell heterogeneity unresolved
- Half-sample is a reproducibility reference, not a mathematical ceiling
- It needs an atlas as reproducible as Tahoe — on L1000 the same estimator loses to the drug mean

A strong baseline — but the atlas's reproducibility is part of the result

For therapeutic discovery: useful prioritization still needs experimental validation